

Generation Interconnection Request System Impact Study Report

For Queue Project Q16-0619

University 500 kV
November 29, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0619. The Interconnection Customer proposes a 1190.6 MW natural gas combined cycle facility interconnecting at the University 500 kV substation in the Indiana area. The request was received on May 17, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 255 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$60,372,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
University 230 kV/1 kV XFMR	Flood St / Cook 500 kV	110.9	1,066	61.1
University 500/1 kV Auto	Flood Street / Cook 500 kV	108.4	1,091	61.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
University 230-1 kV Transformer	Replace transformer with higher-rated unit	\$11,563,000	98% (\$11,348,000)
University 500 kV/1 kV Auto	Replace transformer with higher-rated unit	\$49,953,000	98% (\$49,024,000)
	Total allocated to project		\$60,372,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.